PERSONALISED DIET RECOMMENDATION SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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COMPUTER SCIENCE AND ENGINEERING



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NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "PERSONALISED DIET RECOMMENDATION SYSTEM" is the bonafide work of RAGAVI N (240701407), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, RAGAVI N (240701407), hereby declare that the project report titled "PERSONALISED DIET RECOMMENDATION SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report - including its data model, forms, reports, pages, Deluge scripts, workflows and integrations - was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11 November 2026

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I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

RAGAVI N

ABSTRACT

The growing demand for personalised nutrition guidance highlights the need for efficient systems to manage user information, dietary preferences, allergies, and meal recommendations. Traditional manual methods and spreadsheet-based processes often result in data duplication, calculation errors, time-consuming record management, and difficulty identifying meals that meet individual nutritional requirements. To address these challenges, a Personalised Diet Recommendation System was developed using Zoho Creator, providing a centralised low-code platform for automating diet management activities.

The application consists of modules for Users, Dietary Preferences, Health Conditions, Allergies, Fitness Goals, Meals, Meal Plans, Recommendations, and User Feedback. Zoho Creator forms, reports, and dashboards provide an organised interface for managing information. Deluge scripting and workflows automate BMI and calorie-related calculations, validate recorded allergies, and generate personalised meal recommendations using Zoho GenAI. Integration with Zoho CRM enables user contact synchronisation and duplicate prevention. An OCR-based nutrition label scanning feature extracts text from uploaded food images to support nutritional information collection. Role-based access settings help manage application accessibility.

The application was tested using sample user records, dietary preferences, allergy information, meal entries, and workflow execution logs. CRM synchronisation, duplicate prevention, and OCR configuration were also evaluated. The system centralises diet-related information, reduces repetitive manual data entry, and automates essential calculations and recommendation processes. It demonstrates the potential to improve data consistency, operational efficiency, and accessibility of personalised nutrition guidance while providing a scalable foundation for future enhancements.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Personalised Nutrition; Optical Character Recognition.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Personalised Diet Recommendation System
Application Domain	Healthcare – Nutrition and Diet Management
Student Name & Register No.	Ragavi N – 240701407
Zoho Creator Edition	Trial Edition (Trial currently active)
Application Name (Link Name)	Personalised Diet Recommendation Sytem
Published / Live URL	https://ragavin2024cserajalak208.zohocreatorportal.in
Demo Credentials (per role)	Admin, User
Number of Forms / Subforms	08 / No Subforms
Number of Reports (by type)	List: 04, Kanban: 03
Number of Pages / Dashboards	1 Dashboard
Number of Workflows	Form: 08, Schedule: 03, Approval: 01, Blueprint: 02, Batch: 01
Deluge Functions / Custom Actions	07
Integrations	Zoho CRM (Contacts Synchronization), Zoho GenAI (AI Meal Recommendations), Zoho Creator OCR (Nutrition Label Text Extraction)
Roles & Profiles Defined	Administrator, Standard User, Customer (Portal)
Environments Used	Development → Production
Date of Production Deployment	10 November 2026

TABLE OF CONTENTS

ABSTRACT iv
PROJECT AT A GLANCE v
LIST OF TABLES ix
LIST OF FIGURES x
LIST OF ABBREVIATIONS xi
INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 2
1.6 Organisation of the Report 3
LITERATURE SURVEY AND SYSTEM STUDY 4
2.1 Study of the Existing System 4
2.2 Limitations of the Existing System 5
2.3 Low-Code versus Traditional Development 5
2.4 Literature Survey 6
2.5 Emerging Trends in Low-Code Development 7
2.6 Proposed System 7
REQUIREMENT ANALYSIS 8
3.1 Stakeholders and User Roles 8
3.2 Functional Requirements 8
3.3 Non-Functional Requirements 9
3.4 Hardware and Software Requirements 9
3.5 Use Case Model 10
SYSTEM DESIGN 11
4.1 System Architecture 11
4.2 Data Model 11
4.3 Form Design 13
4.4 Process and Workflow Design 13
4.5 User Interface and Navigation Design 15
4.6 Access Control Design 15
IMPLEMENTATION 16
5.1 Application Setup 16
5.2 Forms and Data Capture 16

5.3 Reports 17
5.4 Pages and Dashboards 18
5.5 Deluge Scripting 19
5.6 Workflows and Automation 20
5.7 Integrations 21
5.8 User Management and Role-Based Access 23
5.9 AI and Analytics Features 23
5.10 Pro-Code Extension 24
5.11 Mobile and Portal Access 24
TESTING AND DEBUGGING 26
6.1 Testing Strategy 26
6.2 Test Cases 26
6.3 Debugging Deluge Scripts 27
6.4 Defect Log 27
6.5 User Acceptance Testing 27
DEPLOYMENT, SECURITY AND GOVERNANCE 28
7.1 Environments and Deployment Strategy 28
7.2 Performance Optimisation 29
7.3 Security Features 29
7.4 Governance and Administration 29
RESULTS AND DISCUSSION 31
8.1 Application Output 31
8.2 Achievement of Objectives 32
8.3 Comparison with the Existing System 33
8.4 Limitations of the Low-Code Approach 33
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 33
CONCLUSION AND FUTURE ENHANCEMENTS 35
9.1 Conclusion 35
9.2 Future Enhancements 35
9.3 Learning Outcomes and Career Relevance 35
REFERENCES 36
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 38
APPENDIX B – COURSE OUTCOME MAPPING 40
APPENDIX C – COMPLETE DELUGE SOURCE CODE 41
APPENDIX D – USER MANUAL 44
D.1 Accessing the Application 44
D.2 Instructions for Admin 44

D.3 Instructions for Standard User 44
APPENDIX E – EVALUATION RUBRIC 46
APPENDIX F – VIVA-VOCE RECORD SHEET.....48

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 5
Table 2.2: Summary of Literature Survey 6
Table 3.1: Stakeholders and User Roles 8
Table 3.2: Functional Requirements 8
Table 3.3: Non-Functional Requirements 9
Table 3.4: Hardware and Software Requirements 9
Table 3.5: Use Case Description – Calculate BMI 10
Table 4.1: Entities and Relationships 12
Table 4.2.1: Field Specification – Users Form 13
Table 4.2.2: Field Specification – Alergies Form 13
Table 4.3: Role–Permission Matrix 15
Table 5.1: Reports Implemented 18
Table 5.2: Deluge Scripts Implemented 19
Table 5.3: Workflows Implemented 21
Table 5.4: Integrations Implemented 22
Table 6.1: Test Cases ..... **Error! Bookmark not defined.**
Table 6.2: Defect Log ..... **Error! Bookmark not defined.**
Table 6.3: User Acceptance Feedback ..... **Error! Bookmark not defined.**
Table 7.1: Deployment Checklist 28
Table 7.2: Performance Optimisation Results 29
Table 7.3: Security Controls Implemented ..... **Error! Bookmark not defined.**
Table 8.1: Mapping of Objectives to Outcomes 32
Table 8.2: Existing System versus Proposed Application ..... **Error! Bookmark not defined.**
Table A.1: Feature Coverage Checklist 39
Table B.1: Mapping of Course Outcomes to Report Chapters 40
Table E.1: Evaluation Rubric for the Individual Project 46
Table E.2: Performance Level Descriptors 47
Table F.1: Viva-Voce Questions and Responses 48
Table F.2: Marks Summary 48

LIST OF FIGURES

Figure 2.1: Process Flow of the Existing Manual Diet Planning System 4
Figure 3.1: Use Case Diagram of the Application 10
Figure 4.1: System Architecture of the Application 11
Figure 4.2: Entity–Relationship Diagram 12
Figure 4.3.1: Workflow / Process Flow for BMI Calculation 14
Figure 4.3.2: Workflow / Process Flow for AI Meal Recommendations 14
Figure 4.3.3: Workflow / Process Flow for Food Label OCR 14
Figure 4.4: Navigation Structure and Wireframe of Personalised Diet Recommendation System 15
Figure 5.1: Application Home and Navigation Sections 16
Figure 5.2: Users Form in the Form Builder 17
Figure 5.3: Users Form in Live Mode 17
Figure 5.4: All Users Report 18
Figure 5.5: Dashboard Page of the Personalised Diet Recommendation System 18
Figure 5.6: AI Meal Recommendations Workflow Configuration 21
Figure 5.7: CRM Contact Synchronisation 23
Figure 5.8: User Management Configuration in Zoho Creator 23
Figure 5.9: OCR Text Extraction 24
Figure 5.10: User Management and Profile Configuration in Zoho Creator 25
Figure 6.1: Debugging and Testing of User Profile Retrieval Function in Zoho Creator ..... **Error! Bookmark not defined.**
Figure 7.1: Development Environment 28
Figure 7.2: Audit Trail 30
Figure 8.1: User Profile Registration and BMI Calculation 31
Figure 8.2: Dietary Preferences and Health Information 31
Figure 8.3: AI-Assisted Meal Recommendations 32

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CRM	Customer Relationship Management
ER	Entity–Relationship
Gemini	Google's Artificial Intelligence Model
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OCR	Optical Character Recognition
OAuth	Open Authorization
PDF	Portable Document Format
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
Zia	Zoho’s AI assistant
Zoho	Zoho Corporation

CHAPTER 1
INTRODUCTION

1.1 Background

The project focuses on the domain of healthcare and nutrition management. It is developed for a hypothetical small-scale nutrition and wellness consultancy that provides personalised dietary guidance and meal-planning services to individuals. The organisation involves administrators, diet consultants, and registered users. The major business activities include maintaining user profiles, recording dietary preferences, managing allergies and health conditions, monitoring fitness goals, and preparing suitable meal recommendations.

Currently, these activities may be performed using spreadsheets, paper records, and separate digital applications. Consultants manually collect user information, calculate nutritional requirements, review food ingredients, and prepare meal plans according to individual needs. As the number of users increases, maintaining accurate and updated records becomes difficult. Therefore, a centralised application is required to simplify information management and support personalised diet planning.

The Personalised Diet Recommendation System is developed using Zoho Creator to automate and organise these activities through interconnected forms, reports, dashboards, Deluge workflows, and AI-based features.

1.2 Problem Statement

The existing diet management process involves manual data entry, separate record maintenance, repetitive nutritional calculations, and time-consuming meal planning. These activities make it difficult to maintain consistent user information, identify suitable meals, and provide personalised dietary recommendations efficiently.

- Data duplication and scattered records across user profiles, dietary preferences, and health information.
- Manual BMI and calorie-related calculations that may result in errors and delays.
- Difficulty checking food allergens and identifying meals suitable for individual dietary requirements.

- Time-consuming preparation of personalised meal recommendations.
- Manual extraction and entry of nutritional information from food labels.

1.3 Objectives

- To design a normalised data model and forms for managing users, dietary preferences, allergies, health conditions, meals, and meal plans.
- To automate BMI and calorie-related calculations using Deluge scripting and workflows.
- To provide role-based dashboards and reports for organised information management.
- To integrate the application with Zoho CRM for user contact synchronisation and duplicate prevention, AI-based meal recommendations using Zoho GenAI, and OCR functionality for extracting text from uploaded nutrition labels.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

The project covers user profile management, dietary preference recording, allergy management, meal planning, nutritional information management, automated calculations, and AI-assisted recommendations. It also includes Zoho CRM integration, OCR-based text extraction, and dashboard-based reporting. The application is developed using Zoho Creator and primarily supports web-based access.

The project does not include medical diagnosis, replacement of professional nutrition consultation, integration with wearable devices, or large-scale commercial deployment. AI recommendations and extracted nutritional information require user verification before being relied upon.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because it provides a low-code environment for developing forms, reports, dashboards, and automated workflows with minimal conventional programming. Its drag-and-drop interface simplifies application development, while Deluge scripting supports custom business logic and automation. The platform also provides integration capabilities with Zoho services, including Zoho CRM and Zoho

GenAI. Built-in hosting, browser-based access, mobile compatibility, and application management features further reduce development and maintenance effort. Therefore, the low-code approach is suitable for developing the proposed application within the available time and resources [1].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, diet planning is mainly performed manually using spreadsheets and personal records. User details, dietary preferences, allergies, and fitness goals are collected and reviewed by a diet consultant. BMI and calorie requirements are calculated manually, and suitable meals are selected after checking their nutritional values and ingredients.

Existing Process:

1. Collect user details.
2. Record dietary preferences and allergies.
3. Calculate BMI and calorie requirements.
4. Review available meals.
5. Check ingredients and allergens.
6. Prepare and share diet recommendations.
7. Maintain user records and feedback.



Figure 2.1: Process Flow of the Existing Manual Diet Planning System

2.2 Limitations of the Existing System

The existing system has the following limitations:

- Manual Data Management: Maintaining records using spreadsheets is time-consuming.
- Calculation Errors: Manual BMI and calorie calculations may lead to errors.
- Allergy Management: Checking food ingredients manually can be difficult.
- Delayed Recommendations: Preparing diet plans requires considerable time.
- Limited Automation: Updating records and tracking feedback involve manual effort.

2.3 Low-Code versus Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–10 weeks	2–3 weeks
Skills required	HTML, CSS, JavaScript, React, backend programming, and database management	Form design, workflow configuration, and basic Deluge scripting
Hosting & infrastructure	Requires separate server setup, domain configuration, and deployment	Application hosted and managed through the Zoho Creator platform
Mobile application	Requires responsive design and additional mobile development if needed	Mobile access supported through the Zoho Creator mobile application
Security & access control	Developers configure authentication, access control, and data protection	Uses platform-provided authentication, permissions, and security controls
Maintenance & change	Developers handle server updates, bug fixes, and application upgrades	Platform handles infrastructure updates; developers maintain application logic and workflows
Customisation limits	Minimal limitations; developers can modify source code, database architecture, and application features as required.	Limited by platform capabilities, supported integrations, available components, and Deluge scripting restrictions.
Cost	Includes developers' effort, hosting, domain, database, and maintenance expenses	Primarily depends on Zoho Creator subscription and additional service usage

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

2.4 Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Bucaioni, A. Cicchetti and F. Ciccozzi (2022)	Modelling in Low-Code Development: A Multi-Vocal Systematic Review	Examines modelling techniques, development approaches, and practices used in low-code platforms.	Supports the selection of Zoho Creator as a low-code application development platform.
2	M. O. Ajimati, N. Carroll and M. Maher (2025)	A Systematic Literature Review and Future Research Agenda	Reviews the adoption, benefits, challenges, and future research directions of low-code and no-code development.	Provides a basis for using low-code development to reduce application development effort.
3	Zoho Corporation	Understand OCR Field – Zoho Creator Help	Explains how OCR extracts text from uploaded images and documents.	Supports the Food Scanner feature for extracting text from food-label images.
4	Zoho Corporation	Creating and Managing Form Workflows – Zoho Creator Help	Describes event-based workflows that automate operations when forms are created, edited, or submitted.	Supports automated calculations and meal recommendation workflows using Deluge scripts.
5	K. Zhao et al. (2026)	Mapping the Landscape of Nutrition Recommendation Systems: Evidence from Peer-Reviewed Research, Apps, and Patents	Reviews existing nutrition recommendation systems, their applications, and developments in personalised nutrition.	Provides research background for generating meal recommendations based on user preferences and nutritional requirements.

Table 2.2: Summary of Literature Survey

2.5 Emerging Trends in Low-Code Development

The project incorporates the following low-code development trends:

- AI-Assisted Development: Zoho GenAI generates meal recommendations.
- Citizen Development: Zoho Creator enables development with minimal coding.
- Hyper-Automation: Deluge scripts automate calculations and workflows.
- Composable Applications: Integration with Zoho CRM and OCR extends application functionality.

2.6 Proposed System

The proposed system uses Zoho Creator to manage user profiles, dietary preferences, allergies, meals, and meal plans. Deluge automates calculations and allergy-based filtering, while Zoho GenAI assists with meal recommendations. CRM integration manages contacts, and OCR extracts text from food labels.

It reduces manual data entry, calculation errors, allergy-checking effort, and delays through centralised records and automated workflows.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages the application.	Manage user records, forms, and application settings.
Diet Consultant	Oversees dietary information and recommendations.	Review user profiles, preferences, allergies, and meal recommendations.
Staff / Operator	Handles application records.	Add and update meals, meal plans, and user information.
Customer / Portal User	Individual using the diet recommendation system.	Submit personal details, record preferences, view recommendations, and provide feedback.

Table 3.1: Stakeholders and User Roles

3.2 Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to enter and maintain personal details, including height, weight, age, and gender.	Administrator, Customer	High
FR-02	The system shall calculate BMI using the user's height and weight.	Customer, Diet Consultant	High
FR-03	The system shall calculate an estimated calorie target based on available user profile information and activity level.	Customer, Diet Consultant	High
FR-04	The system shall allow users to record dietary preferences, health conditions, allergies, and fitness goals.	Customer, Diet Consultant	High
FR-05	The system shall generate meal recommendations using Zoho GenAI.	Customer, Diet Consultant	High
FR-06	The system shall check recorded allergies against meal ingredients before recommending meals.	Customer, Diet Consultant	High
FR-07	The system shall extract text from uploaded food-label images using OCR.	Customer, Staff	Medium

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall synchronise user contact information with Zoho CRM Contacts.	Administrator	Medium

Table 3.2: Functional Requirements

3.3 Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports should load within 5 seconds under normal network conditions.
NFR-02	Security	User information shall be accessible only to authorised users.
NFR-03	Usability	The application shall provide simple forms and navigation through a web browser.
NFR-04	Availability	The application shall be accessible through Zoho Creator when the platform and internet connection are available.
NFR-05	Maintainability	Forms, workflows, and Deluge scripts shall be modifiable when requirements change.
NFR-06	Auditability	Record creation and modification details should be traceable using available platform audit features.

Table 3.3: Non-Functional Requirements

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho CRM and Zoho GenAI
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

Table 3.4: Hardware and Software Requirements

3.5 Use Case Model

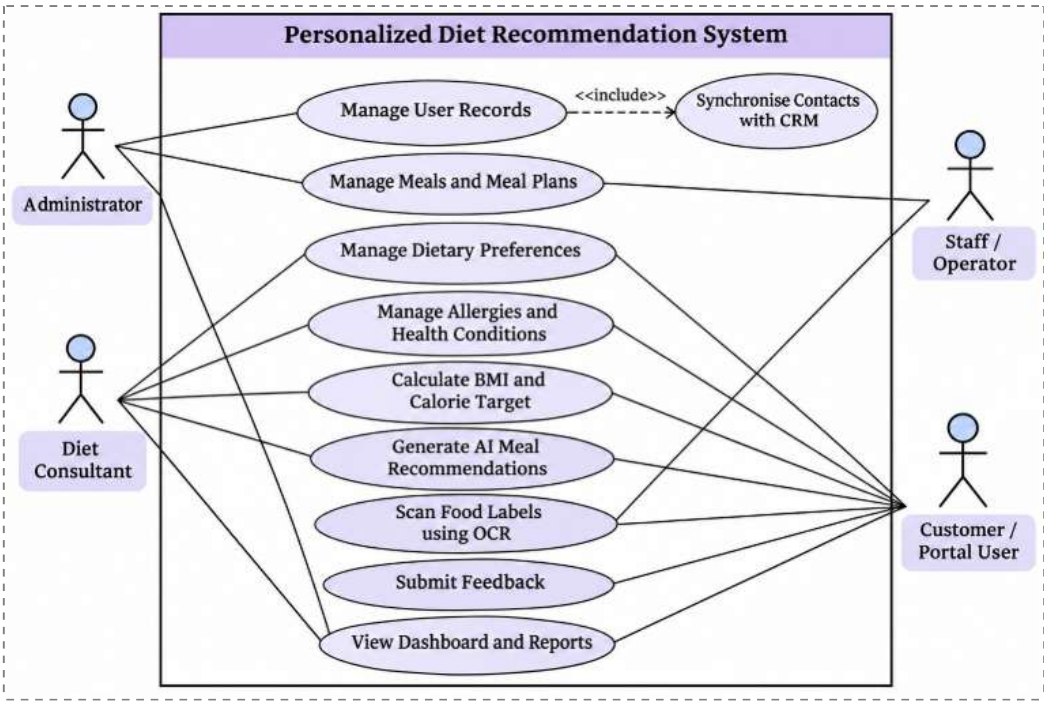


Figure 3.1: Use Case Diagram of the Application

Use Case ID	UC-01
Use Case Name	Calculate BMI
Actor(s)	Customer / Portal User, Diet Consultant
Pre-condition	User profile must contain valid height and weight values.
Main Flow	1. User enters height and weight details. 2. The application validates the entered values. 3. The Deluge workflow calculates BMI using the entered measurements. 4. The calculated BMI is assigned to the user record.
Alternate Flow	If height or weight is missing or invalid, BMI calculation is skipped.
Post-condition	BMI value is calculated and made available in the user form.
Related FR	FR-01, FR-02

Table 3.5: Use Case Description – Calculate BMI

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

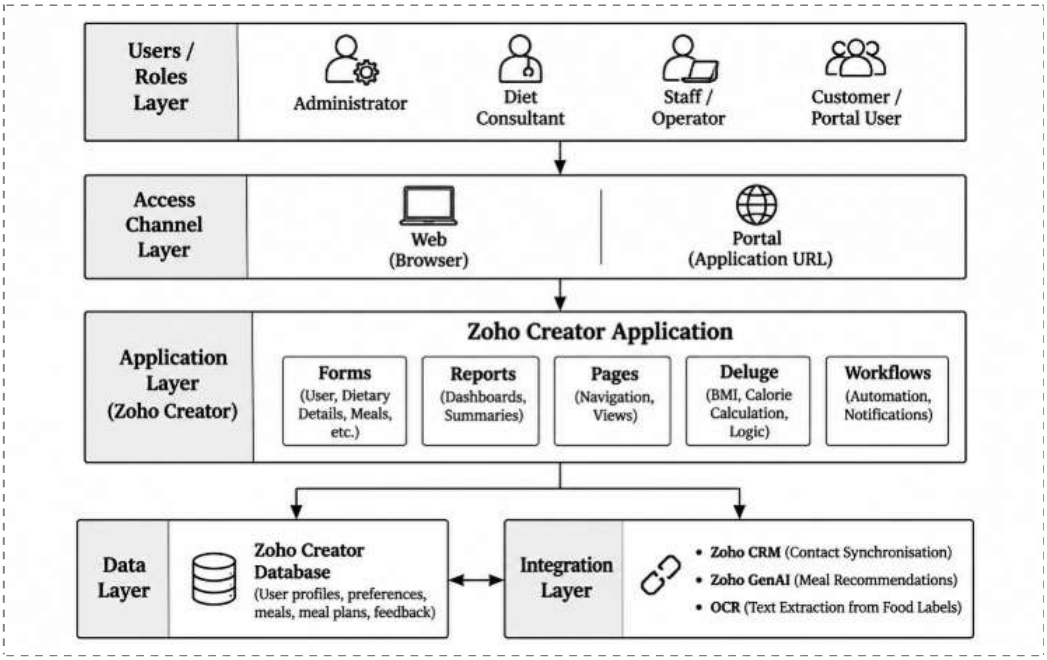


Figure 4.1: System Architecture of the Application

The system follows a five-layer architecture consisting of the Users/Roles, Access Channel, Application, Data, and Integration layers. The Users/Roles Layer includes the Administrator, Diet Consultant, Staff/Operator, and Customer, while the Access Channel Layer provides access through a web browser and Zoho Creator portal.

The Application Layer uses Zoho Creator forms, reports, pages, Deluge scripts, and workflows to manage user details, dietary preferences, calculations, and meal recommendations. The Data Layer stores user profiles, meal details, preferences, meal plans, and feedback. The Integration Layer connects Zoho CRM for contact synchronisation, Zoho GenAI for meal recommendations, and OCR for extracting text from food label images. The Integration Layer connects Zoho CRM for contact synchronisation, Zoho GenAI for meal recommendations, and OCR for extracting text from food label images.

4.2 Data Model

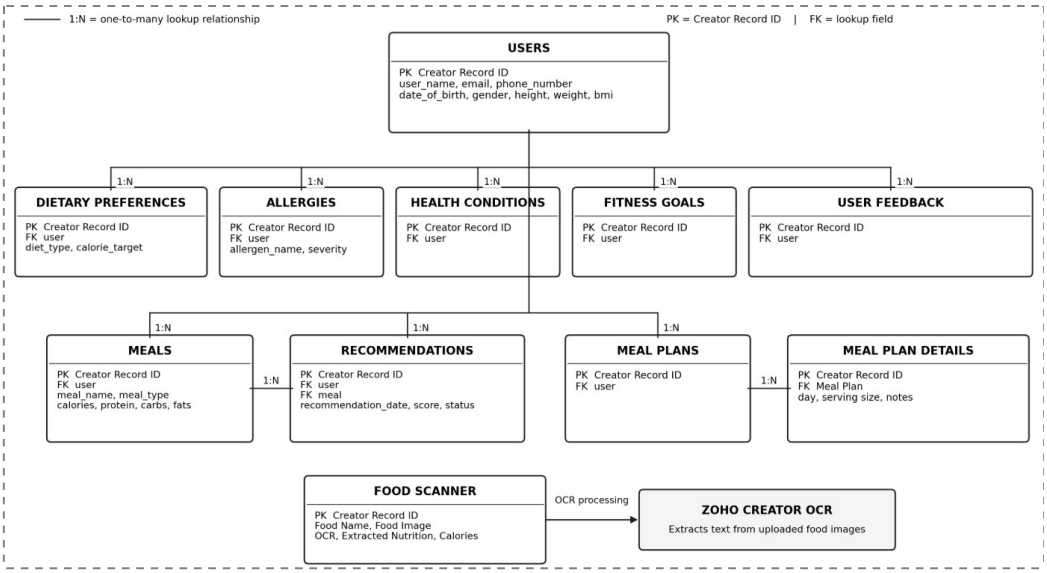


Figure 4.2: Entity–Relationship Diagram

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Users	User Name, Email, Gender, Height, Weight	Dietary Preferences, Allergies, Health Conditions, Fitness Goals, User Feedback, Meals, Recommendations, Meal Plans	One-to-Many	Lookup fields
Dietary Preferences	User, Diet Type, Calorie Target	Users	Many-to-One	Lookup
Allergies	User, Allergen Name, Severity	Users	Many-to-one	Lookup
Health Conditions	User, Health Condition	Users	Many-to-one	Lookup

Table 4.1: Entities and Relationships

4.3 Form Design

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
User Name	user_name	Single Line	Yes	Not specified	Stores user's name
Email Address	email	Email	Yes	Email format	Stores email address
Phone Number	phone_number	Phone	Yes	No duplicate values	Contact information
Date of Birth	date_of_birth	Date	Not specified	Not specified	Stores date of birth
Gender	Gender	Dropdown	Yes	Not specified	Gender selection
Height	Height	Number	Yes	Positive value	Used for BMI calculation
Weight	weight	Number	Yes	Positive value	Used for BMI calculation

Table 4.2.1: Field Specification – Users Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
User	user	Lookup	Yes	Linked to users	Associate allergy with a user
Allergen name	Allergen_name	Single line	Yes	Not specified	Records allergen
Reaction Type	Reaction_type	Dropdown	Yes	Not specified	Records reaction type
Description	Description	Multiline	Not specified	Not specified	Additional allergy information
Severity	Severity	Dropdown	Yes	Not specified	Indicates allergy severity

Table 4.2.2: Field Specification – Allergies Form

4.4 Process and Workflow Design

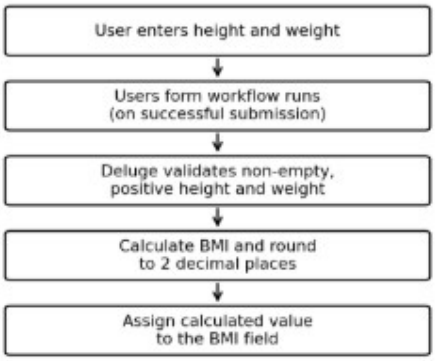


Figure 4.3.1: Workflow / Process Flow for BMI Calculation

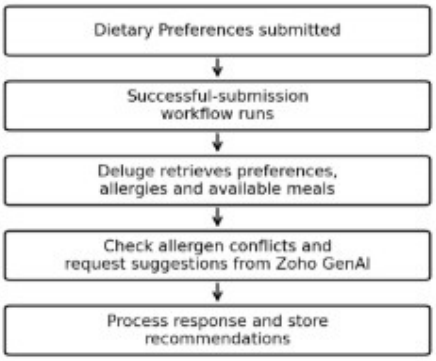


Figure 4.3.2: Workflow / Process Flow for AI Meal Recommendations

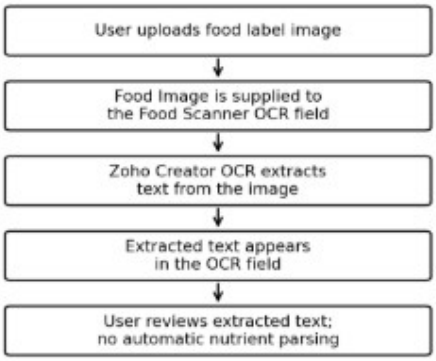


Figure 4.3.3: Workflow / Process Flow for Food Label OCR

4.5 User Interface and Navigation Design

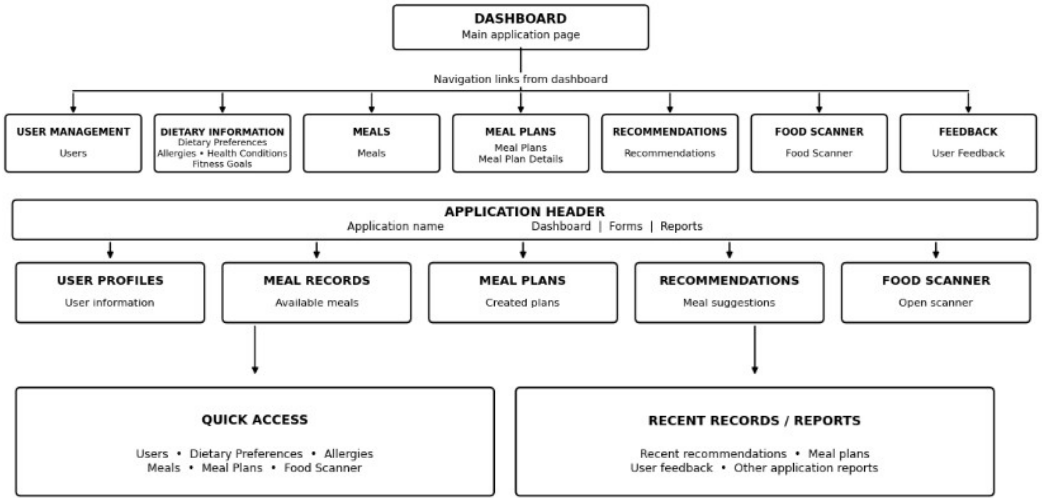


Figure 4.4: Navigation Structure and Wireframe of Personalised Diet Recommendation System

4.6 Access Control Design

Component	Administrator	Manager	Staff	Portal User
Form: Users	A E V D	V	A E V	A E V (own)
Form: Dietary Preferences	A E V D	A E V	A E V	A E V (own)
Form: Allergies	A E V D	A E V	A E V	A E V (own)
Form: Health Conditions	A E V D	A E V	A E V	A E V (own)
Form: Fitness Goals	A E V D	A E V	A E V	A E V (own)
Form: Meals	A E V D	A E V	A E V	V
Form: Meal Plans	A E V D	A E V	A E V	V (own)
Form: Meal Plan Details	A E V D	A E V	A E V	V (own)
Form: Recommendations	A E V D	A E V	A E V	V (own)
Form: Food Scanner	A E V D	A E V	A E V	A E V (own)
Form: User Feedback	A E V D	V	A E V	A E V (own)
Page: Dashboard	V	V	V	V (own)
Reports: Application Records	V	V	V	—

Table 4.3: Role–Permission Matrix

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Personalised Diet Recommendation System was developed using Zoho Creator Education Edition. The application, named **Dietary Recommendation Platform**, contains 11 forms and a Dashboard page for managing user profiles, dietary preferences, allergies, meals, meal plans, recommendations and feedback. Deluge workflows and Zoho integrations support automated operations.

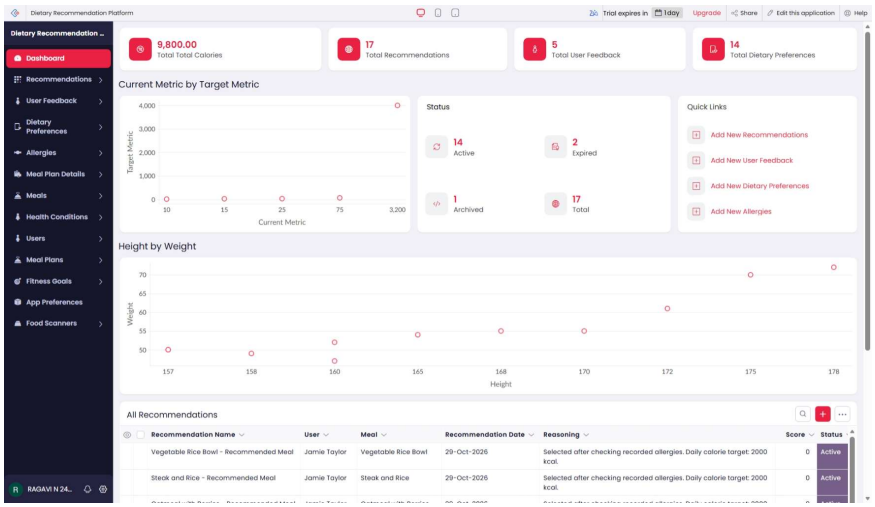


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The application contains 11 forms for capturing and managing user profiles, dietary preferences, allergies, health conditions, fitness goals, meals, meal plans, recommendations, feedback and food-label information. Fields use appropriate data types such as Single Line, Number, Date, Dropdown, Lookup and Image. Lookup fields connect related records, while Deluge workflows support automated calculations and data processing.

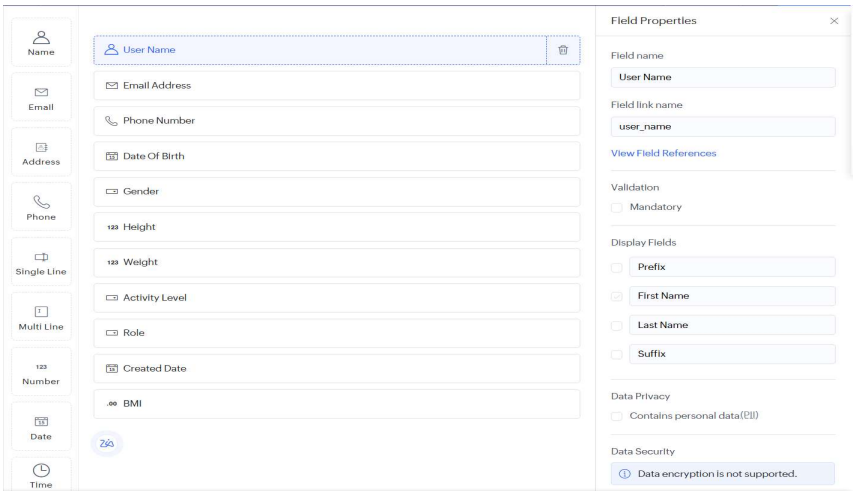


Figure 5.2: Users Form in the Form Builder

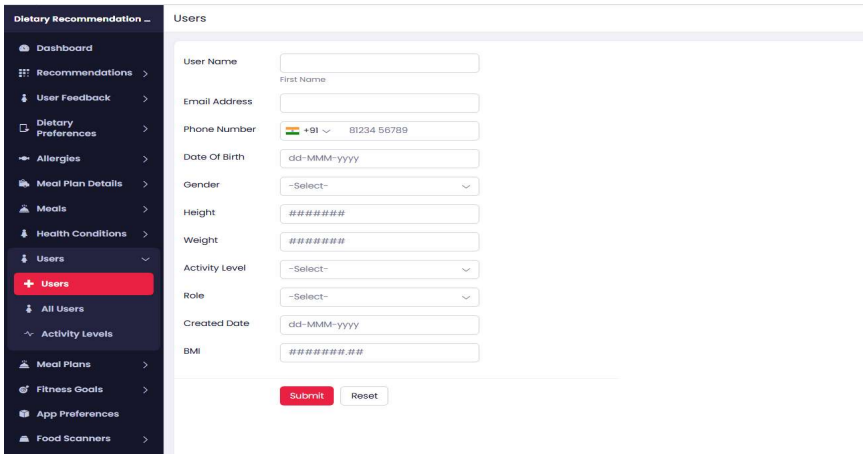


Figure 5.3: Users Form in Live Mode

5.3 Reports

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Users Report	List	Users	Default settings	Default actions	Not separately configured
Dietary Preferences Report	List	Dietary Preferences	Default settings	Default actions	Not separately configured
Allergies Report	List	Allergies	Default settings	Default actions	Not separately configured

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Health Conditions Report	List	Health Conditions	Default settings	Default actions	Not separately configured
Users Report	List	Users	Default settings	Default actions	Not separately configured

Table 5.1: Reports Implemented

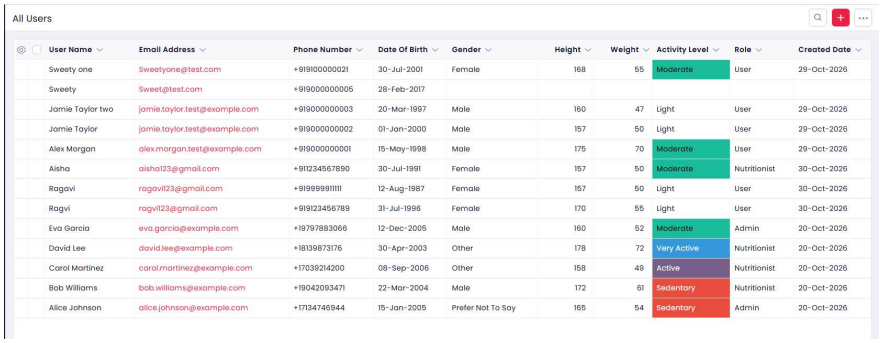


Figure 5.4: All Users Report

5.4 Pages and Dashboards

The application contains a Dashboard page that serves as the main entry point for accessing its modules. It provides navigation to user profiles, dietary information, meals, meal plans, recommendations, food scanning and feedback. The page is intended for application users to access the available functions from a central location.

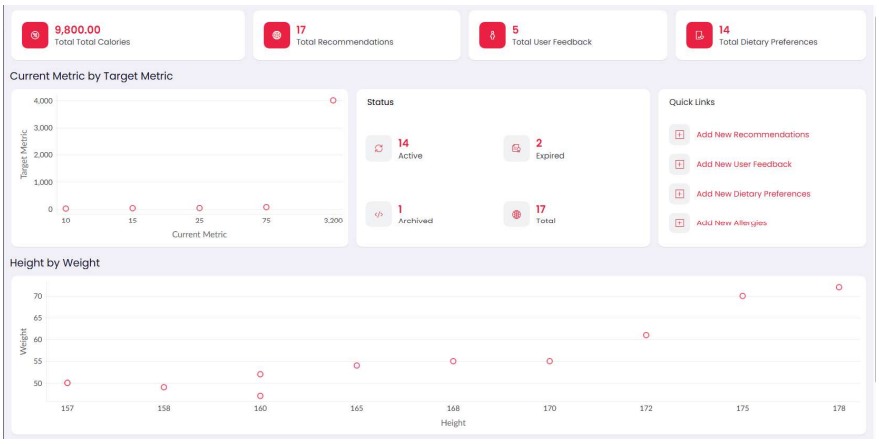


Figure 5.5: Dashboard Page of the Personalised Diet Recommendation System

5.5 Deluge Scripting

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Users → Successful Form Submission	Calculate BMI using height and weight	Conditional statements, arithmetic operations, null checks, rounding
DS-02	Dietary Preferences → On Load of Form	Estimate and populate the calorie target	Record retrieval, conditional statements, arithmetic operations, rounding
DS-03	Dietary Preferences → Successful Form Submission	Generate AI-assisted meal recommendations using Zoho GenAI	Zia integration, data retrieval, conditional statements, response processing

Table 5.2: Deluge Scripts Implemented

Code 5.1: BMI Calculation Script

```
// DS-01 | Form: Users | Event: Successful Submission
// Check whether height and weight are available
if(input.height != null && input.weight != null)
{
    if(input.height > 0 && input.weight > 0)
    {
        // Calculate BMI using height in metres
        bmiValue = input.weight / ((input.height / 100) *
(input.height / 100));

        // Store BMI rounded to two decimal places
        input.bmi = bmiValue.round(2);
    }
}
```

- DS-01 – BMI Calculation: Checks whether valid height and weight values are available, calculates BMI and assigns the rounded result to the BMI field.

- DS-02 – Calorie Target: Retrieves the logged-in user's profile, uses weight, height and activity level to estimate a calorie target, and populates the corresponding field. The current formula uses a fixed age assumption of 30 and a male BMR equation.

- DS-03 – AI Meal Recommendations: Retrieves dietary preferences, allergy information and available meal records. It uses Zoho GenAI to assist in. generating meal recommendations, which are intended to be stored in the Recommendations form.

Code 5.2: Calorie Target Estimation Script

```
// DS-02 | Form: Dietary Preferences | Event: On Load
// Retrieve the logged-in user's profile
loginEmail = ifnull(zoho.loginuserid,"");
calorieTarget = 2000;

if(loginEmail != "")
{
    userProfile = users[email == loginEmail];

    if(userProfile.count() > 0)
    {
        userWeight = userProfile.weight;
        userHeight = userProfile.height;
        activityLevel = userProfile.activity_level;

        // Estimate calorie target using available profile details
        if(userWeight != null && userHeight != null)
        {
            bmr = 10 * userWeight + 6.25 * userHeight - 5 * 30 + 5;

            if(activityLevel == "Very Active")
                { calorieTarget = bmr * 1.9; }
            else if(activityLevel == "Active")
                { calorieTarget = bmr * 1.55; }
            else if(activityLevel == "Moderate")
                { calorieTarget = bmr * 1.375; }
            else
                { calorieTarget = bmr * 1.2; }
        }
    }
}

input.calorie_target = calorieTarget.round(0);
```

5.6 Workflows and Automation

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
BMI Calculation	Form Workflow	Users – Successful Form Submission	Height and weight are available and greater than zero	Calculate and assign BMI	DS-01
Calorie Target Estimation	Form Workflow	Dietary Preferences – On Load of Form	Logged-in user profile and required details are available	Estimate and populate calorie target	DS-02

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
AI Meal Recommendations	Form Workflow	Dietary Preferences – Successful Form Submission	Dietary preferences and relevant meal data are available	Retrieve dietary information, check recorded allergies and request AI-assisted recommendations	DS-03
CRM Contact Synchronisation	Form Workflow	Form event – To be verified	Check whether a matching CRM contact exists	Create a contact in Zoho CRM if no duplicate is found	CRM Integration Script
BMI Calculation	Form Workflow	Users – Successful Form Submission	Height and weight are available and greater than zero	Calculate and assign BMI	DS-01

Table 5.3: Workflows Implemented

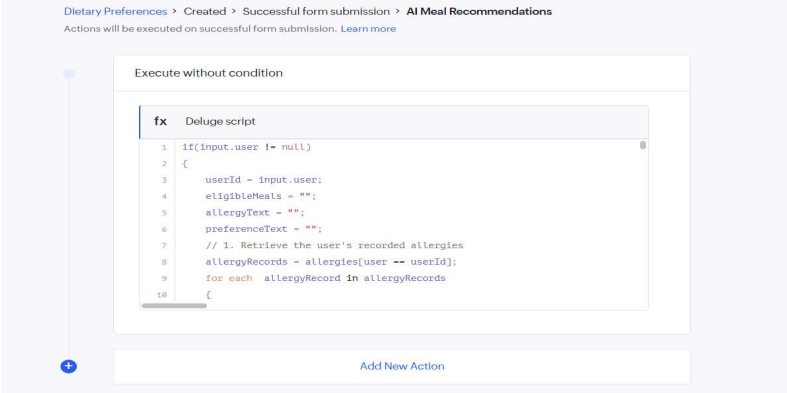


Figure 5.6: AI Meal Recommendations Workflow Configuration

5.7 Integrations

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
CRM Synchronisation	Zoho CRM – Contacts	Zoho CRM integration through Deluge workflow	Zoho authorised connection (verify authentication details)	Transfers user contact information and checks for duplicate contacts

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
AI Meal Recommendations	Zoho GenAI	Deluge Zia task	Zoho Creator platform authorisation	Sends dietary context and receives AI-assisted meal recommendation responses
Food Label Text Extraction	Zoho Creator – OCR	Built-in OCR field	Zoho Creator platform service	Extracts text from uploaded food-label images

Table 5.4: Integrations Implemented

Code 5.3: Zoho CRM integration script

```
contactEmail = ifnull(input.email,"").trim();
if(contactEmail != "")
{
	existingContacts =
zoho.crm.searchRecords("Contacts","(Email:equals:" + contactEmail +
")",1,200,Map(),"zoho_crm_connection");
	if(existingContacts == null || existingContacts.isEmpty())
	{
		contactData = Map();
		fullName = ifnull(input.user_name,"").trim();
		nameParts = fullName.toList(" ");
		if(nameParts.size() > 1)
		{
			contactData.put("First_Name",nameParts.get(0));

	contactData.put("Last_Name",fullName.remove(nameParts.get(0) + "
"));
		}
		else
		{
			contactData.put("Last_Name",fullName);
		}
		contactData.put("Email",contactEmail);
		if(input.phone_number != null && input.phone_number != "")
		{
			contactData.put("Phone",input.phone_number);
		}
		response =
zoho.crm.createRecord("Contacts",contactData,Map(),"zoho_crm_connection");
		info response;
	}
	else
	{
		info "Contact already exists for email: " + contactEmail;
	}
}
else
{
	info "CRM sync skipped because the email address is blank.";
}
```

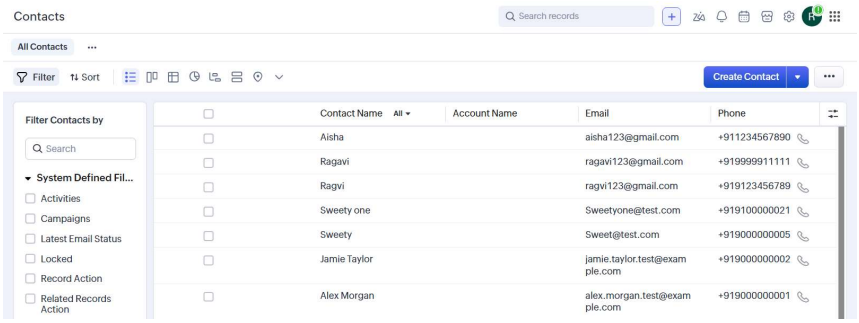


Figure 5.7: CRM Contact Synchronisation

5.8 User Management and Role-Based Access

The application includes a Users form with a role field to maintain user role information. The proposed role-based permissions are documented in Chapter 4; separate permission profiles and record-level access rules have not yet been confirmed as configured.

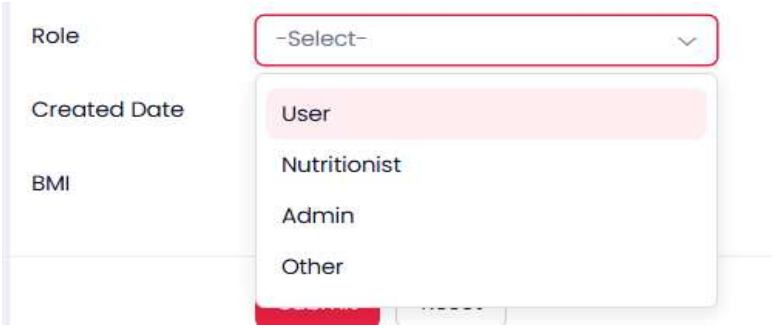


Figure 5.8: User Management Configuration in Zoho Creator

5.9 AI and Analytics Features

The application uses Zoho GenAI for AI-assisted meal recommendations and Zoho Creator OCR for extracting text from food-label images. Both features were tested using sample inputs.

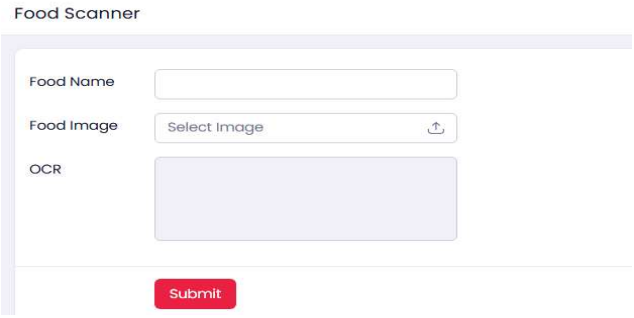


Figure 5.9: OCR Text Extraction

5.10 Pro-Code Extension

The application was developed using Zoho Creator's low-code environment and does not currently include external pro-code extensions such as Zoho Catalyst, JavaScript widgets or Creator REST APIs. This approach helped simplify development and reduce the need for extensive programming. In future enhancements, these technologies could be used to build customised interfaces, integrate external services and implement advanced application features.

5.11 Mobile and Portal Access

The Personalized Diet Recommendation System is developed using Zoho Creator, which allows users to access the application through a web browser. Users can log in with their registered Zoho account and access the features available according to their assigned roles and permissions.

5.11.1 Mobile Access

Users can access the application through a mobile device by following these steps:

1. Open a web browser on a smartphone.
2. Navigate to the application URL provided by the administrator.
3. Log in using the registered Zoho account credentials.
4. Open the Personalized Diet Recommendation System application.
5. Access the My Diet Dashboard to view diet recommendations and meal-plan information.
6. Use the Food Scanner feature to upload food-label images, if enabled for the account.

5.11.2 User Access and Permissions

The application supports two main user roles:

- Admin: Manages meal information, reviews generated meal plans and monitors application records.
- Standard User: Accesses the dashboard, requests personalized diet recommendations and uses available food-scanning features.

Access to forms, reports and application features depends on the permissions assigned to each role.

5.11.3 Portal Access

Users access the application using their registered Zoho accounts and the application URL shared by the administrator. The available features are controlled through the configured user permissions.

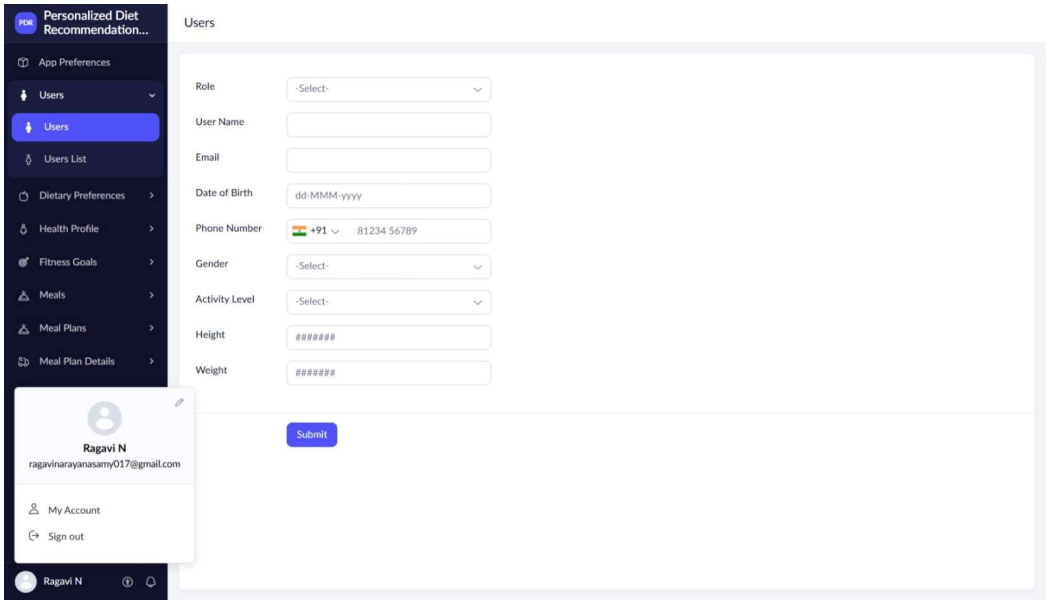


Figure 5.10: User Management and Profile Configuration in Zoho Creator

Result: Mobile-browser access and role-based application access provide users with a convenient way to interact with the diet recommendation system**.**

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out to verify the functionality of forms, Deluge scripts and automated workflows. Sample records were used to check BMI calculation, calorie target estimation, AI-assisted meal recommendations, OCR text extraction and Zoho CRM contact synchronisation. Workflow outputs and execution logs were reviewed to identify errors. Role-based access testing and formal user acceptance testing were not conducted.

6.2 Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit valid user details	Enter valid name, email, phone and other profile details.	User record is created successfully.	User record created successfully.	Pass
TC-02	FR-01	Submit user details with invalid email	Enter an incorrectly formatted email address.	Email validation error is displayed.	Email validation error was displayed.	Pass
TC-03	FR-02	Calculate BMI with valid values	Enter height and weight and submit the Users form.	BMI is calculated and rounded to two decimal places.	Calculation log message was observed.	Pass
TC-04	FR-02	Calculate BMI with missing values	Leave height or weight empty and submit.	BMI calculation is skipped.	BMI calculation skipped due to missing values.	Pass

Table 6.1: Test Cases

6.3 Debugging Deluge Scripts

Debugging was performed using Deluge info statements, workflow execution logs and error messages to identify issues in scripts. One issue occurred in the BMI calculation workflow when height or weight values were empty, causing calculation errors. Null checks and positive-value validations were added before performing the calculation. The workflow was then tested using sample user records.

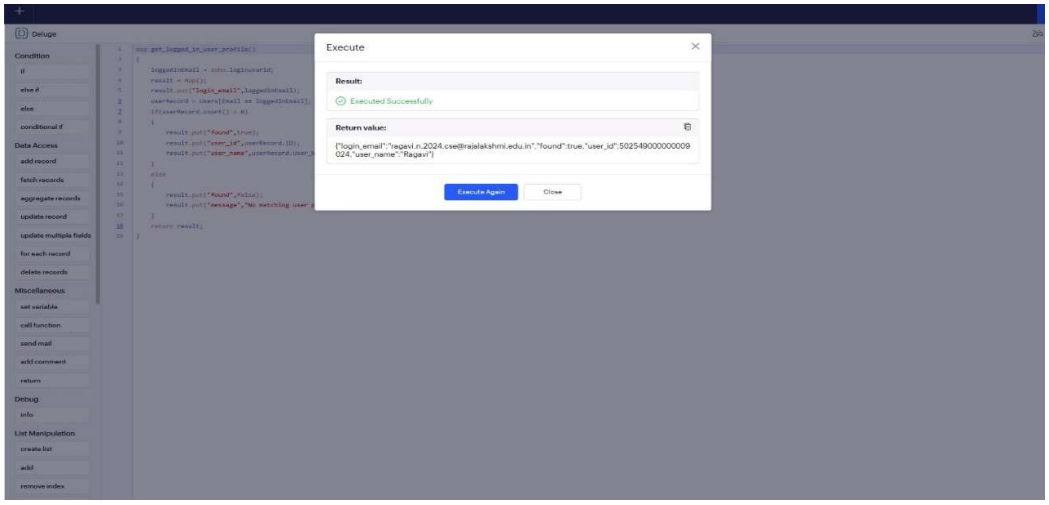


Figure 6.1: Debugging and Testing of User Profile Retrieval Function in Zoho Creator

6.4 Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	BMI calculation failed when height or weight was empty.	Null values were used in arithmetic operations.	Added null checks and positive-value validation before calculating BMI.	Closed

Table 6.2: Defect Log

6.5 User Acceptance Testing

User	Role Tested	Feedback	Action Taken
Alice Johnson	To be confirmed	Requested a healthy dessert recipe and shared a hydration reminder.	Feedback recorded for future enhancements.
Bob Williams	To be confirmed	Suggested adding more vegetables to daily meals.	Feedback recorded for future enhancements.

Table 6.3: User Acceptance Feedback

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Zoho Creator development environment. Separate Stage and Production environments were not used, and the application has not been published to production. Changes were tested within the development environment before being retained. Version notes and a formal rollback procedure were not configured. These can be introduced in future deployment stages to support controlled releases and recovery from errors.

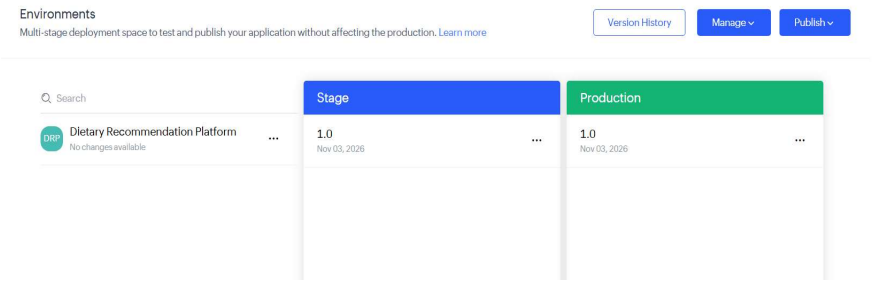


Figure 7.1: Development Environment

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

Table 7.1: Deployment Checklist

7.2 Performance Optimisation

Optimisation Applied	Component	Before	After	Improvement
Criteria-based record retrieval	DS-02 – Calorie Target Workflow	General record retrieval	User profile retrieved using email criteria	More targeted data retrieval
Null checks before calculations	DS-01 – BMI Calculation	Empty values could cause calculation errors	Validates height and weight before calculation	Improved calculation reliability

Table 7.2: Performance Optimisation Results

7.3 Security Features

Security Control	Where Applied	Threat Mitigated
Zoho Creator platform authentication	Application access	Unauthorised application access
Authorised Zoho connection	Zoho CRM integration	Exposure of integration credentials
Input validation	Users form and Deluge workflows	Invalid or incomplete data entry
Duplicate contact checking	CRM synchronisation workflow	Duplicate contact records

Table 7.3: Security Controls Implemented

7.4 Governance and Administration

The application is maintained through the Zoho Creator workspace. Forms, Deluge scripts and workflows can be updated as required. The application uses consistent form and field names to organise its components. Formal backup schedules, audit-trail reviews and licence-management procedures have not been established. Maintenance and future updates will be handled by the application developer.

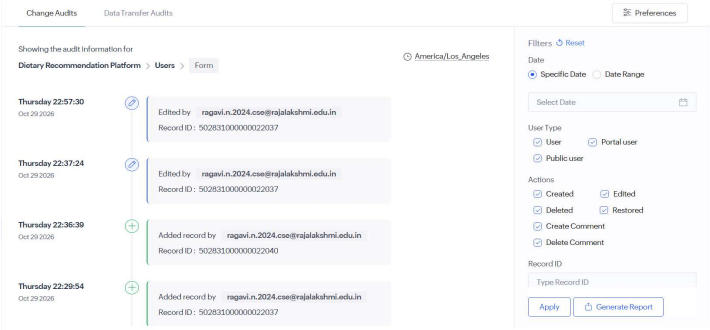


Figure 7.2: Audit Trail

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A sample user scenario is used to demonstrate the application's workflow. The user first enters personal details, followed by dietary preferences and allergy information. The system calculates the BMI and estimated calorie target using Deluge workflows. Based on the available dietary information, Zoho GenAI assists in generating meal recommendations. The following screenshots illustrate the major stages of the application.

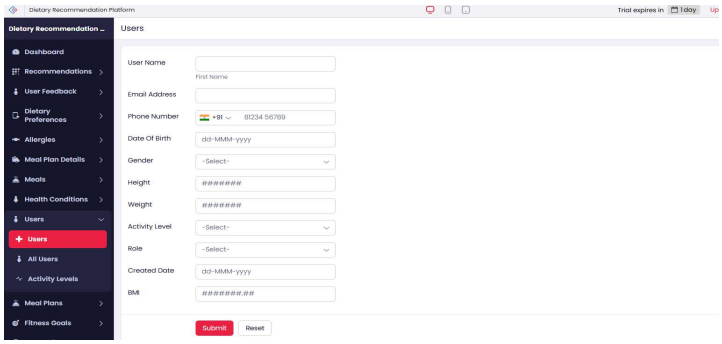


Figure 8.1: User Profile Registration and BMI Calculation

The user enters personal information such as name, email, date of birth, height and weight through the Users form. The BMI calculation workflow processes the available height and weight values.

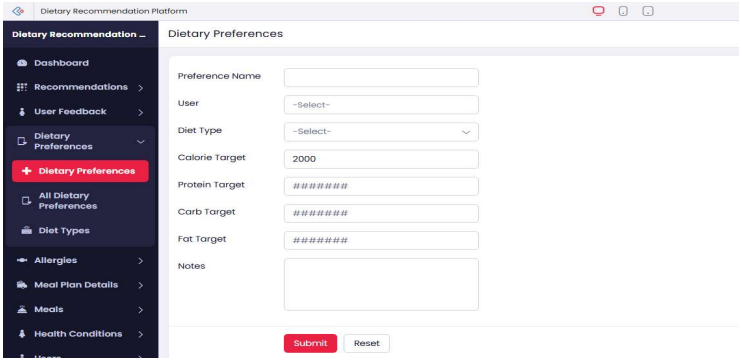


Figure 8.2: Dietary Preferences and Health Information

The user records dietary preferences, allergies, health conditions and fitness goals. The calorie target is estimated based on the available profile information.

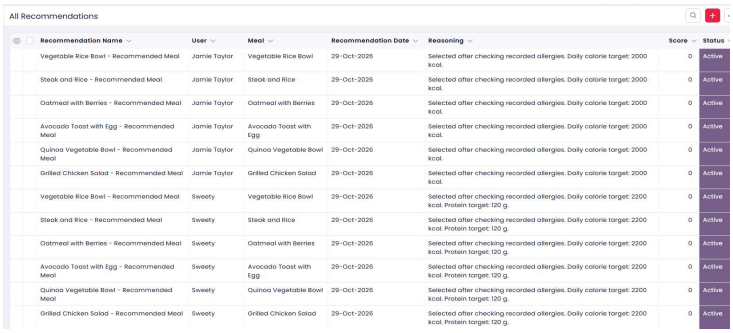


Figure 8.3: AI-Assisted Meal Recommendations

The submitted dietary information is processed through the recommendation workflow. Zoho GenAI assists in generating suitable meal recommendations, which are stored in the Recommendations form.

8.2 Achievement of Objectives

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 1: Provide personalised nutrition recommendations based on dietary preferences, health conditions and nutritional requirements.	Sections 5.2, 5.5, 5.6 and 5.9	User profiles, dietary information and AI-assisted meal recommendations are supported through Zoho Creator and Zoho GenAI.
Objective 2: Integrate nutrition and fitness planning to support individual wellness goals.	Sections 5.2 and 5.4	Dietary preferences, fitness goals and meal plans can be maintained through application forms. Automated workout scheduling has not been implemented.
Objective 3: Monitor user progress and continuously improve recommendations.	Sections 5.2 and 5.3	User records and feedback can be maintained. Automated progress monitoring and adaptive recommendation refinement remain future enhancements.

Table 8.1: Mapping of Objectives to Outcomes

8.3 Comparison with the Existing System

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Manual collection of user details, calculations and meal planning.	Centralised forms and automated Deluge workflows.	Reduces repetitive manual effort.
Manual steps	Preferences, allergies and nutritional information are managed manually.	Forms, lookup fields and automated workflows manage the information.	Reduces manual data handling.
Report generation	Records are maintained and reviewed manually.	Default reports are automatically available for application forms.	Provides organised access to stored records.
Data accessibility	Information may be scattered across spreadsheets and separate records.	Information is maintained centrally in Zoho Creator.	Improves centralised data access.

Table 8.2: Existing System versus Proposed Application

8.4 Limitations of the Low-Code Approach

The development of the application using Zoho Creator involved certain limitations. The platform provides predefined UI components, which restrict extensive interface customisation. Deluge scripting also has limitations when implementing complex logic and debugging workflows. Additionally, AI and OCR features depend on platform availability and usage limits. These challenges were addressed by using built-in components, dividing operations into separate workflows and validating scripts through execution logs. The application also depends on the Zoho ecosystem, which may create vendor dependency and make migration to other platforms more difficult.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The application demonstrates all three levels of the No-Code → Low-Code → Pro-Code spectrum. No-code configuration was used to create forms, reports, lookup relationships and the Dashboard page. Low-code development was used to implement Deluge scripts and workflows for BMI calculation, calorie target estimation, AI-required in the current implementation. However, technologies such as Zoho Catalyst, JavaScript widgets and external APIs could be introduced for advanced customisation and complex functionality. Moving to pro-code would be beneficial if the application's requirements exceed the capabilities of Zoho Creator

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Personalised Diet Recommendation System was successfully developed using Zoho Creator Education Edition. The application provides a centralised platform for managing user profiles, dietary preferences, allergies, health conditions, fitness goals, meals, meal plans, recommendations and user feedback. Deluge workflows were implemented to support BMI calculation and calorie target estimation, while Zoho GenAI was integrated to assist with meal recommendations. The application also incorporates Zoho CRM contact synchronisation and Zoho Creator OCR for food-label text extraction.

The project demonstrates how low-code development can simplify application development through forms, lookup relationships, workflows and platform integrations. The implemented features help organise dietary information and reduce repetitive manual tasks. Overall, the project achieves its objective of developing a centralised dietary recommendation platform, while providing a foundation for further improvements in automation, access control and recommendation accuracy.

9.2 Future Enhancements

- Predictive Analytics: Incorporate advanced analytics to identify dietary patterns, monitor user progress and provide more personalised nutrition recommendations.

- Payment Gateway Integration: Integrate a secure payment gateway to support subscription-based plans and premium nutrition services.

- Zoho Catalyst Integration: Develop Catalyst-based microservices to handle complex data processing and advanced recommendation logic beyond the capabilities of the current low-code implementation.

9.3 Learning Outcomes and Career Relevance

The development of the Personalized Diet Recommendation System provided practical experience in low-code application development using Zoho Creator. I gained knowledge of form creation, data modelling, lookup relationships, Deluge scripting, workflow automation and integration with services such as Zoho CRM and Zoho GenAI. I also learned how to configure OCR functionality, validate user inputs, debug scripts and organise application data. This project helped me understand how low-code platforms simplify application development while still allowing custom logic through scripting. These skills are relevant to several career pathways in the low-code development ecosystem. The experience gained can support a career as a **Citizen Developer** by enabling the development of business applications using visual tools; a **Low-Code Developer** by combining platform components with scripting; a **Solutions Consultant** by understanding requirements and translating them into application workflows; or an **Automation Engineer** by designing processes that reduce repetitive manual tasks. Overall, the project strengthened my understanding of low-code development and its role in building practical business solutions.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Zoho Creator Application	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Users, Meals, Meal Plans	5.2	Fig. 5.2
II	Lookup field (relationship between forms)	Y	Users and Meal Plans	4.2, 5.2	Fig. 5.3
II	Subform (parent–child data)	N	Separate child form used	4.2, 5.2	—
II	Field validations / layout rules	Y	Users Form	5.2	Code 5.1
II	List report with filter, grouping, custom action	Y	Meal Plans Report	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Meal Plans Kanban Report	5.3	Fig. 5.5
II	Pivot table / chart report	N	Meal Plans Chart Report	5.3	—
II	Page / dashboard built with UI Builder	Y	My Diet Dashboard	5.4	Fig. 5.7
II	Deluge – form events (validate / success / field actions)	Y	Users / Food Scanner Workflow	5.5	Code 5.2
II	Deluge – functions, fetch, loops, collections	Y	Meal Plan Generation Functions	5.5	Code 5.3
II	Email / in-app notifications	Y	Meal Plan Approval / Rejection	5.5, 5.6	Fig. 5.8
II	Form workflow	Y	Food Scanner	5.6	Code 5.4
II	Scheduled workflow	Y	Expired Meal Plans	5.6	Fig. 5.9
II	Approval process	Y	Meal Plan Review	5.6	Fig. 5.10
II	Blueprint	Y	Meal Plan Approval Lifecycle	5.6	Fig. 5.11

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Batch / report workflow	Y	Meal Plans Report	5.6	Code 5.5
II	Testing & debugging evidence	Y	Test Cases and Execution Results	6	Table 6.1
III	Integration with a Zoho service	Y	Zoho CRM Integration	5.7	Code 5.6
III	Connection / REST API call (invokeurl)	N	Gemini AI Integration	5.7	—
III	Webhook or Creator REST API	N	Not configured	5.7	—
III	Users, roles and permission profiles	Y	Admin and Standard User Roles	5.8	Fig. 5.12
III	Record-level access criteria / customer portal	N	Own-record access not verified	5.8	—
III	Zia / AI feature	N	Gemini AI Meal Plan and Food Scanner	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Meal Plan Chart Report	5.9	Fig. 5.14
IV	Performance optimisation applied & measured	N	Performance measurements pending	7.2	—
IV	Deployment across environments	Y	Deployment Configuration	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	N	Complete security verification pending	7.3	—
IV	Governance (audit trail, administration)	Y	Creator Audit Trail	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	Y	Custom Deluge Functions	5.10	Code 5.8
V	Low-code limitations & spectrum analysis	Y	Application Analysis	8.4, 8.5	Table 8.1
—	Mobile app / portal access	Y	Zoho Creator Portal	5.11	Fig. 5.15

Table A.1: Feature Coverage Checklist

Total features implemented: 24 / 32

APPENDIX B – COURSE OUTCOME MAPPING

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

Table B.1: Mapping of Course Outcomes to Report Chapters

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 AI Diet Recommendation – validate_gemini_response

```
// Script ID : DS-01
// Location  : My Diet Dashboard → Recommend Meals for Me button
// Purpose   : Executes the AI diet recommendation process and submits
//             the generated meal plan for admin review.
string validate_gemini_response(){
      p = thisapp.get_logged_in_user_profile();
      if(p.get("found") != true) return "User not found.";
      u = p.get("user_id");
      h = Health_Profile[User == u];
      if(h.count() == 0) return "Health Profile missing.";
      r = thisapp.generate_ai_meal_plan();
      try{
d =
r.toMap().get("responseText").get("candidates").get(0).get("content")
.get("parts").get(0).get("text").toMap();}
      catch(e) { return "Invalid Gemini response."; }
      meals = d.get("recommended_meals");
      if(meals == null) return "No meals generated.";
      for each m in meals{
            a = Meals[ID == m.get("meal_id").toLong()];
            if(a.count() == 0 || a.Availability_Status !=
"Available")
                  return "Invalid meal.";}
      diet = Dietary_Preferences[User == u];
      fit = Fitness_Goals[User == u];
      if(diet.count() == 0 || fit.count() == 0) return "Preferences
missing.";
      s = zoho.currentdate;
      pid = insert into Meal_Plans
            Plan_Name = "AI Meal Plan - " + s
            User = u
            Plan_Description = d.get("plan_explanation")
            Plan_Type = "AI Generated"
            Fitness_Goal = fit.Primary_Fitness_Goal
            Start_Date = s
            End_Date = s
            Plan_Status = "Pending Review"
            Added_User = zoho.loginuser];
      for each m in meals{
            id = m.get("meal_id").toLong();
            a = Meals[ID == id];
            insert into Meal_Plan_Details[
                  Meal_Plan = pid
                  Meals = id
                  User = u
                  Scheduled_Date = s
                  Meal_Type = a.Meal_Category
                  Serving_Quantity = 1];}
      return "SUCCESS! Plan ID: " + pid;}
```

C.2 CRM Contact Creation

```
// Script ID : DS-02
// Location  : Zoho Creator → Deluge Function / CRM Integration
// Purpose   : Creates a contact record in Zoho CRM using the
//             Zoho_CRM_Connection connection.

sync_user_to_crm: string sync_user_to_crm(int userID)
{
      userRecord = Users[ID == userID];
      if(userRecord.count() == 0)
      {
            return "Error: User not found in Creator.";
      }
      if(ifnull(userRecord.Email,"") == "" || ifnull(userRecord.User_
Name,"") == "")
      {
            return "Error: User name or email is missing.";
      }
      searchResponse = zoho.crm.v8.searchRecords("Contacts","(Email:
equals:" + userRecord.Email + ")",1,200,Map(),"zoho_crm_connection");
      contactData = Map();
      contactData.put("Last_Name",userRecord.User_Name);
      contactData.put("Email",userRecord.Email);
      contactData.put("Phone",userRecord.Phone_Number);
      if(searchResponse.size() > 0)
      {
            crmContactID = searchResponse.get(0).get("id");
            updateResponse =
zoho.crm.v8.updateRecord("Contacts",crmContactID.toLong(),contactData,
Map(),"zoho_crm_connection");
            return "CRM Contact updated: " +
updateResponse.toString();
      }
      else
      {
            createResponse =
zoho.crm.v8.createRecord("Contacts",contactData,Map(),"zoho_crm_
connection");
            return "CRM Contact creation response: " +
createResponse.toString();
      }
}
```

C.3 Food Scanner – OCR and AI Analysis

```
// Script ID : DS-03
// Location  : Food Scanner → Form Workflow
// Purpose   : Extracts text from uploaded food-label images and
//             stores the extracted text in the OCR field.

string validate_gemini_response()
{
	u = thisapp.get_logged_in_user_profile().get("user_id");
	d = thisapp.generate_ai_meal_plan().toMap();
	meals = d.get("recommended_meals");
	if(u == null || meals == null) return "Validation Failed.";
	for each m in meals
		if(Meals[ID == m.get("meal_id").toLong()].count() == 0)
			return "Invalid Meal.";
	pid = insert into Meal_Plans [User=u, Plan_Type="AI
Generated"];
	for each m in meals
		insert into Meal_Plan_Details [Meal_Plan=pid,
Meals=m.get("meal_id"), User=u];
	return "SUCCESS! Plan created: " + pid;
}
```

C.4 User Profile Retrieval

```
// Script ID : DS-04
// Location  : Zoho Creator → Custom Function
// Purpose   : Retrieves the logged-in user's profile information.

get_logged_in_user_profile: map get_logged_in_user_profile()
{
	loggedInEmail = zoho.loginuserid;
	result = Map();
	result.put("login_email",loggedInEmail);
	userRecord = Users[Email == loggedInEmail];
	if(userRecord.count() > 0)
	{
		result.put("found",true);
		result.put("user_id",userRecord.ID);
		result.put("user_name",userRecord.User_Name);
	}
	else
	{
		result.put("found",false);
		result.put("message","No matching user profile found.");
	}
	return result;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open a web browser and navigate to the Zoho Creator application link shared by the administrator.

2. Log in using your registered Zoho account credentials.

3. After successful login, open the Personalized Diet Recommendation System application.

4. The application dashboard will appear with the features available to your assigned role.

5. On a mobile device, open the application link in a mobile browser and log in using the same credentials.

D.2 Instructions for Admin

- Step 1: Log in to the application using your administrator account.
- Step 2: Open the Meals form to add or update meal details, including food name and nutritional information.
- Step 3: Open the Food Scanner to upload a food-label image and view the extracted text and AI analysis.
- Step 4: Open the Meal Plans report to view AI-generated diet plans submitted by users.
- Step 5: Review pending meal plans and update their approval status.
- Step 6: Use the available reports to monitor users, meal records, food scans and generated meal plans.

D.3 Instructions for Standard User

- Step 1: Log in to the application using your registered Zoho account.
- Step 2: Open My Diet Dashboard to view your diet-related information and meal-plan status.

- Step 3: Select Recommend Meals for Me to generate a personalized diet recommendation based on your health profile, dietary preferences and fitness goal.
- Step 4: Wait for the success message confirming that your meal plan has been submitted for admin review.
- Step 5: View your meal-plan information and check its status on the dashboard or in the available meal-plan report.
- Step 6: Use the Food Scanner to upload a food-label image and view the extracted text and AI-generated nutritional analysis, if this feature is enabled for your account.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Ragavi N	Register No.	2116240701407
Project Title	Personalized Diet Recommendation System	Date	11-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Allotted
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Ragavi N	Register No.	2116240701407
Department / Section	Computer Science and Engineering	Date of Viva	11-11-2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____